

Institute of Leather Engineering and Technology
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TANNERY WASTEWATER TREATMENT

Study Material for Undergraduate Students

CHAPTER 2

SECONDARY TREATMENT

Physicochemical and Advanced Processes

Filtration · Adsorption · Chemical Oxidation · Ion Exchange

Membrane Processes · Evaporation and ZLD · Disinfection

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A NOTE ON TERMINOLOGY

In much of the wastewater literature the word 'secondary treatment' refers specifically to BIOLOGICAL treatment, and the physicochemical processes described in this chapter are called 'tertiary' or 'advanced' treatment.

In this study material we group the technologies by their treatment FUNCTION rather than by their traditional position in the train. This chapter therefore covers the non-biological removal of dissolved and colloidal pollutants, and Chapter 3 covers biological treatment. Where a process is conventionally called tertiary, this is noted in the text so that you recognise the term in other books.

2.1 Introduction: What Primary Treatment Left Behind

Chapter 1 ended with a clarified effluent. Screening removed the coarse matter, equalisation steadied the flow, sulphide was oxidised, chromium was recovered, coagulation and settling removed the bulk of the suspended solids. That is a great deal of progress, and the water leaving the primary clarifier looks far cleaner than the raw effluent.

It is nevertheless a long way from being dischargeable. Four categories of pollutant survive primary treatment, and it is these that the present chapter and the next must address.

What survives	Why primary treatment missed it	Dealt with in
Dissolved biodegradable organic matter	It is dissolved — nothing to screen or settle	Chapter 3 (biological)
Refractory organic matter and colour	Not biodegradable, and not settleable	This chapter (2.5, 2.6)
Dissolved salts (TDS)	Truly dissolved; no physical or biological process removes them	This chapter (2.9, 2.10)
Residual fine solids and colloids	Too small to settle in the time available	This chapter (2.4)
Residual metals	A small soluble fraction escapes recovery	This chapter (2.7, 2.8)
Pathogens	Not affected by physical separation	This chapter (2.11)

Table 2.1 Pollutants surviving primary treatment, and where each is addressed.

The first row belongs to Chapter 3 and is not discussed here. Everything else in that table is the subject of this chapter.

KEY IDEA

Primary treatment removes what can be screened, floated and settled. Biological treatment removes what is biodegradable. The processes in this chapter remove what is neither — the refractory organics, the colour, the dissolved salts and the residual fine solids that no other stage can touch.

2.1.1 The tannery's characteristic difficulty

For a tannery, one entry in Table 2.1 dominates all the others. Recall from Chapter 1 that raw tannery effluent carries 10,000–40,000 mg/L of dissolved solids, against a discharge limit of 2,100 mg/L. Salt is used lavishly in this industry — for curing hides, in pickling, in chrome tanning — and all of it ends in the effluent.

No unit operation in Chapter 1 reduced the TDS by a single milligram, and no biological process in Chapter 3 will do so either. Salt is chemically inert, fully dissolved, and invisible to every conventional treatment step. Only two answers exist: reduce the salt at source, which belongs to cleaner production; or separate it physically by a membrane, which is Section 2.9.

2.2 Classification and the Order of the Units

2.2.1 Classification by mechanism

Mechanism	What it does	Processes
Separation by size	Retains particles or molecules larger than an opening	Filtration; microfiltration; ultrafiltration
Separation by affinity	Pollutant sticks to, or exchanges onto, a solid	Adsorption on carbon; ion exchange
Chemical destruction	The pollutant molecule is broken apart	Ozonation; advanced oxidation; chlorination
Chemical conversion	The pollutant is converted to an insoluble solid	Metal precipitation
Separation by solution–diffusion	Water passes through a dense film; solutes do not	Nanofiltration; reverse osmosis
Phase change	Water is evaporated, leaving solids behind	Evaporation; crystallisation; distillation

Table 2.3 Classification of secondary (physicochemical) processes by mechanism.

2.3.2 The order in which the units are placed

These processes are almost always arranged in a fixed order, and the reason is that each protects the next.

Filtration → Adsorption or Oxidation → Membranes → Disinfection

and, where zero liquid discharge is required, evaporation and crystallisation after the membranes

The logic is as follows.

1. **Filtration comes first.** Suspended solids blind carbon beds, foul membranes and shield micro-organisms from disinfectants. Every later unit fails prematurely if fine solids are still present.
2. **Adsorption or oxidation comes second.** Colour and refractory organic matter would otherwise foul membranes organically and consume chlorine or ozone chemically.
3. **Membranes come third.** Now fed with clean, low-colour water, they can remove the dissolved salts without rapid fouling.
4. **Disinfection comes last,** acting on water that is already clear, so that only a small dose is required.

KEY IDEA

Every unit in this chapter depends on the one before it. Placing a carbon bed or a membrane ahead of filtration does not save money — it destroys the carbon or the membrane. The sequence is not a convention; it is a physical necessity.

2.4 Filtration

Purpose

To remove the fine suspended solids that escape the clarifier, producing a clear water suitable for the units that follow.

Principle

Water is passed through a bed of granular medium. Particles are retained by three mechanisms acting together, and it is important to understand that straining is only one of them:

- **Straining** — particles larger than the pore openings are held at the surface.
- **Sedimentation and interception** — within the bed, particles settle onto the grains or collide with them as the water follows a tortuous path between them.
- **Adhesion** — having touched a grain, particles are held there by surface forces; a thin biological or chemical film on the grains assists this.

Because the second and third mechanisms operate throughout the depth of the bed, a filter removes particles very much smaller than its pore openings.

Process description

Water enters at the top of the bed and percolates downward, leaving through an underdrain. As solids accumulate, the head loss across the bed rises and the filtrate quality eventually deteriorates. The bed is then BACKWASHED: clean water, often assisted by air scour, is driven upward at a velocity sufficient to fluidise the bed, lifting the accumulated solids clear and carrying them to waste. Backwash water amounts to 2–5% of throughput and is returned to the head of the works.

Types of filter

Type	Description	Application
Rapid gravity sand filter	Sand bed 0.6–0.9 m deep; 5–15 m ³ /m ² ·h	The standard workhorse
Dual or multimedia filter	Anthracite over sand (over garnet)	Longer runs, higher solids capacity
Pressure filter	Sand in a closed steel vessel under pressure	Small plants; pre-treatment for RO
Disc or cloth-media filter	Rotating cloth-covered discs	Very compact; retrofit where space is short
Slow sand filter	Fine sand at 0.1–0.2 m ³ /m ² ·h	Very high quality; large area

Table 2.4 Types of filter.

Why dual media performs better

This point is worth understanding rather than memorising. In a single sand bed, backwashing sorts the grains so that the FINEST settle at the top. All the solids are therefore caught in the first few centimetres, the surface blinds quickly, and the head loss rises fast — while the rest of the bed does almost no work.

Placing coarse, light anthracite above fine, dense sand reverses the gradation in the direction of flow. Water now meets coarse media first, solids penetrate deeper, and the whole depth of the bed is used. Filter runs are consequently much longer for the same quality of filtrate. After backwashing the two media re-sort themselves correctly, because the anthracite is lighter even though its grains are larger.

Important operating parameters

Parameter	Typical value	Comment
Filtration rate	5 – 15 m ³ /m ² ·h	Higher rates need deeper or coarser beds
Bed depth	0.6 – 1.2 m	Deeper beds allow higher rates
Effective grain size	0.5 – 1.2 mm (sand)	Finer gives better quality, shorter runs
Run length	12 – 48 h	Ended by head loss or breakthrough
Backwash rate	25 – 60 m ³ /m ² ·h	Must fluidise the bed
Backwash water	2 – 5% of throughput	Returned to the plant inlet

Pollutants removed

Residual suspended solids, typically from 30–80 mg/L down to below 10 mg/L; with them, the BOD and COD they carry; and much of the residual turbidity. Filtration removes no dissolved matter whatever.

Advantages and limitations

Advantages	Limitations
Simple, reliable and well understood	Removes no dissolved pollutants
Produces the clear water every later unit needs	Requires backwashing and a backwash water supply
Moderate capital and operating cost	Backwash water must be treated again
Also removes some BOD and COD with the solids	Blinds rapidly if the feed is poorly clarified

Application to tannery wastewater

Filtration is the foundation of the whole advanced train and is rarely omitted. In a tannery it is placed after the clarifier (or after biological treatment) and before any carbon bed, membrane or disinfection unit. Because tannery effluent carries a high solids load, dual-media or pressure filters are usually preferred to plain sand, and the backwash frequency is higher than in municipal practice.

2.5 Adsorption on Activated Carbon

Purpose

To remove dissolved colour and refractory (non-biodegradable) organic matter — the pollutants that neither settling nor biology can remove.

Principle

Adsorption — the adhesion of dissolved molecules onto the SURFACE of a solid. The dissolved substance removed is the ADSORBATE; the solid it adheres to is the ADSORBENT.

Adsorption must be distinguished carefully from absorption, a word that looks similar and means something quite different. In ADsorption the molecules stick to the surface, like dust settling on a wall. In ABSorption they penetrate into the bulk of the material, like water soaking into a sponge. Carbon treatment is adsorption.

Two forces are involved:

- **Physical adsorption (physisorption)** — weak van der Waals attraction. It is reversible, which is what allows the carbon to be regenerated. This dominates in water treatment.
- **Chemical adsorption (chemisorption)** — actual chemical bonding to the surface; stronger, but largely irreversible.

Why activated carbon works

Activated carbon is produced by charring a carbon-rich material — coconut shell, wood, coal, and in principle leather waste — and then ACTIVATING it with steam or chemicals at high temperature. Activation burns out a vast internal network of pores. Three properties result:

Property	Significance
Internal surface area 500 – 1,500 m²/g	A single teaspoonful has roughly the surface area of a football pitch. This is where adsorption occurs
Macropores and micropores	Macropores act as highways carrying molecules inward; micropores (< 2 nm) provide the adsorption sites
Largely non-polar surface	Strongly attracts large, non-polar organic molecules — exactly what dyes, tannins and syntans are

The third property explains why carbon is so effective on colour specifically. Dye and tannin molecules are large, aromatic and poorly soluble in water — precisely the molecules that adsorb most strongly.

Forms of carbon and process description

Property	Powdered (PAC)	Granular (GAC)
Particle size	< 0.1 mm	0.6 – 4 mm
How applied	Dosed into the water, mixed, then filtered off	Packed in a fixed-bed column; water flows through
Contact	Batch; 30 – 60 min	Continuous; empty-bed contact 10 – 30 min
Regeneration	Usually not regenerated	Thermally regenerated and reused
Typical use	Occasional or seasonal colour problems	Continuous polishing

In a GAC column the water enters at the top and passes down through the bed. The carbon nearest the inlet saturates first, and a MASS TRANSFER ZONE — the band in which adsorption is actually occurring

— travels slowly down the column. When that zone reaches the outlet, pollutant begins to appear in the treated water. This moment is called **BREAKTHROUGH**, and the carbon must then be regenerated or replaced.

The adsorption isotherm

The capacity of a carbon for a given pollutant is described at constant temperature by an **ISOTHERM**. The Freundlich equation is the one normally used for wastewater:

$$q_e = K^f \cdot C_e^{(1/n)}$$

q_e = mass adsorbed per unit mass of carbon (mg/g); C_e = equilibrium concentration remaining in solution (mg/L); K^f and n are constants

Taking logarithms gives a straight line, which is how the constants are found experimentally:

$$\log q_e = \log K^f + (1/n) \log C_e$$

K^f indicates the **CAPACITY** of the carbon — a higher value means more pollutant held per gram. The exponent $1/n$ indicates the **INTENSITY** of adsorption; a value between 0.1 and 1 signifies favourable adsorption. The isotherm is determined by a simple bench test in which a fixed volume of the wastewater is contacted with different masses of carbon.

WORKED EXAMPLE 2.1 — Carbon capacity

A bench test on a tannery effluent gives the following result for one carbon dose:

Initial COD C_0 = 400 mg/L; final COD C_e = 130 mg/L; carbon dose 1.00 g in 500 mL.

Calculate the adsorption capacity and the removal achieved.

Step 1 COD removed = 400 – 130 = 270 mg/L

Step 2 Mass removed = 270 mg/L × 0.5 L = 135 mg

Step 3 Adsorption capacity q_e = 135 mg / 1.00 g = 135 mg COD per g of carbon

Step 4 Removal = (400 – 130) / 400 × 100 = 67.5%

COMMENT: knowing q_e , the carbon consumption of a full-scale plant can be estimated. Treating 400 m³/d and removing the same 270 mg/L would require 400 × 270 / 135 = 800 kg of carbon per day if used once through — which is why granular carbon, regenerated and reused, is preferred at scale.

Important operating parameters

Parameter	Typical value	Comment
Empty-bed contact time	10 – 30 min	The key design parameter for GAC
Bed depth	1.5 – 3 m	Deeper beds give a longer service life
Hydraulic loading	5 – 15 m ³ /m ² ·h	As for a filter
PAC dose	20 – 200 mg/L	Determined by jar test
pH	Slightly acidic to neutral favoured	Adsorption of organics improves at lower pH
Feed suspended solids	< 10 mg/L	Higher values blind the carbon

Advantages and limitations

Advantages	Limitations
Very effective on colour and refractory COD	Capacity is finite; breakthrough is inevitable
Simple to operate; no complex chemistry	Carbon is costly; regeneration needs a furnace
Also removes odour and some organic micro-pollutants	Does not remove dissolved salts at all
Reliable and predictable performance	Requires well-filtered feed water
Regeneration recovers most of the capacity	The pollution is transferred, not destroyed

KEY IDEA

Adsorption TRANSFERS pollution; it does not destroy it. The colour and COD removed from the water are now held on the carbon, and that spent carbon must be regenerated or disposed of as a waste. Oxidation processes, by contrast, genuinely destroy the molecule — which is the essential difference between Sections 2.5 and 2.6.

Application to tannery wastewater

Tannery effluent is strongly coloured by dyes, syntans and vegetable tannins, and carries a substantial refractory COD which biological treatment leaves untouched. Activated carbon is therefore a standard polishing step where colour and COD limits must be met. It is placed after filtration and, usually, after biological treatment — so that the readily biodegradable load has already been removed and the carbon is used only on the fraction that needs it. Using carbon on unfiltered or biologically untreated effluent wastes it very rapidly.

2.6 Chemical Oxidation and Advanced Oxidation Processes

Purpose

To DESTROY refractory organic molecules and colour by chemical oxidation, rather than merely transferring them to another phase.

Principle

Oxidation — a reaction in which a substance loses electrons. In water treatment an oxidising agent is added which takes electrons from the pollutant molecule, breaking it apart and converting it — ideally — into carbon dioxide and water.

Oxidising agents differ greatly in strength, and the strength determines what they can attack. The standard measure is the oxidation potential:

Oxidant	Oxidation potential (V)	Comment
Hydroxyl radical ($\cdot\text{OH}$)	2.80	The most powerful practical oxidant
Ozone (O_3)	2.07	Strong; generated on site
Hydrogen peroxide (H_2O_2)	1.78	Convenient liquid
Potassium permanganate	1.68	Used for specific duties
Chlorine	1.36	Cheap; principally a disinfectant
Oxygen	1.23	Slow without a catalyst

Table 2.5 Oxidation potentials of common oxidants.

2.6.1 Ozonation

Ozone (O_3) is an unstable triatomic form of oxygen. It cannot be stored and must be generated on site by passing dry air or oxygen through a high-voltage electrical discharge:



Yields are 1–3% ozone from air and 6–14% from pure oxygen. The gas is dissolved into the wastewater in a contact column, usually through fine-bubble diffusers at the base of a deep tank, with 10–20 minutes of contact. Because ozone is toxic, the off-gas must be passed through a thermal or catalytic destructor before release.

Ozone attacks pollutants by two routes:

- **Direct attack** — the ozone molecule reacts selectively with unsaturated bonds, the carbon–carbon double bonds of dyes and aromatic compounds. This cleavage is why the process is sometimes called OZONOLYSIS, and it is why ozone destroys colour so effectively: the double bonds it attacks are exactly the chromophores that make a dye molecule coloured.
- **Indirect attack** — at higher pH, ozone decomposes in water to form hydroxyl radicals, which are stronger still and attack almost any organic molecule without selectivity.

2.6.2 Advanced oxidation processes (AOPs)

Advanced oxidation process — any process designed to generate hydroxyl radicals ($\cdot\text{OH}$) in sufficient quantity to mineralise refractory organic pollutants.

Process	How radicals are generated	Notes
O₃ / H₂O₂ (peroxone)	Peroxide accelerates ozone decomposition	Widely used; no residue
UV / H₂O₂	UV light splits H ₂ O ₂ into two ·OH	Effective; needs clear water
Fenton reaction	$\text{H}_2\text{O}_2 + \text{Fe}^{2+} \rightarrow \cdot\text{OH} + \text{OH}^- + \text{Fe}^{3+}$	Simple, cheap, effective on coloured effluent; produces iron sludge
Photo-Fenton	Fenton with UV light	Faster; regenerates the catalyst
Electrochemical oxidation	Oxidants generated at electrodes	Compact; electrode cost

Table 2.6 Advanced oxidation processes.

The Fenton process deserves particular mention for tannery work. It requires only hydrogen peroxide and a ferrous salt, both cheap and easily handled, works well at the acidic pH of about 3 that tannery effluent can easily be adjusted to, and is very effective against colour. Its drawback is the iron sludge produced, which must be settled and disposed of.

A valuable secondary benefit

Oxidation seldom mineralises a molecule completely at an economic dose. What it does instead is **BREAK LARGE MOLECULES INTO SMALLER FRAGMENTS** — and those fragments are frequently biodegradable, even though the parent molecule was not.

This suggests a powerful arrangement: place a modest oxidation stage **BETWEEN** two biological stages. The first biological stage removes the easily biodegradable matter; the oxidation stage converts the refractory residue into biodegradable fragments; the second biological stage then consumes them cheaply. The combination treats what neither process could handle alone, and it uses far less oxidant than complete mineralisation would require.

Important operating parameters

Parameter	Typical value	Comment
Ozone dose — disinfection	5 – 15 mg/L	On clean effluent
Ozone dose — colour removal	20 – 50 mg/L	Often the governing duty in tanneries
Ozone dose — refractory COD	30 – 100+ mg/L	Cost rises steeply with COD
Contact time	10 – 20 min	In a deep contact column
Transfer efficiency	85 – 95%	Deep tanks transfer better
Energy for generation	8 – 12 kWh/kg O ₃ (oxygen feed)	15–20 kWh/kg from air
Fenton pH	≈ 3	The optimum for radical generation

WORKED EXAMPLE 2.2 — Ozone requirement for colour removal

A tannery discharges 500 m³/d of treated effluent whose colour must be reduced to meet the 150 Hazen limit. Bench trials show 35 mg/L is required. Transfer efficiency is 90%; the generator is oxygen-fed at 10 kWh/kg.

Step 1 Ozone required = 0.035 kg/m³ × 500 m³/d = 17.5 kg O₃/d

Step 2 Generator output = 17.5 / 0.90 = 19.4 kg/d, say 20 kg/d (0.83 kg/h)

Step 3 Power = 0.83 kg/h × 10 kWh/kg ≈ 8.3 kW

Parameter	Typical value	Comment
Step 4 Daily energy = $20 \times 10 = 200$ kWh/d ANSWER: a generator of about 20 kg/d capacity, drawing roughly 8–10 kW and consuming about 200 kWh per day. COMMENT: compare this with Worked Example 2.1, where carbon removed COD by adsorption. The design question is always which is cheaper for the particular effluent — the electricity for ozone, or the carbon and its regeneration.		

Advantages and limitations

Advantages	Limitations
DESTROYS the pollutant — no residue to dispose of	High capital and energy cost
Very effective on colour and refractory COD	Ozone must be generated on site
Ozone reverts to oxygen; adds dissolved oxygen	No residual protection
Raises the biodegradability of what remains	Bromate may form if bromide is present
No chlorinated by-products	Toxic gas; off-gas destructor essential
Also disinfects	Requires skilled operation

Adsorption or oxidation? A comparison

Aspect	Activated carbon	Ozone / AOP
Mechanism	Adsorption — transfers the pollutant	Oxidation — destroys the molecule
Residue	Spent carbon to regenerate or dispose of	None; ozone reverts to oxygen
Capacity	Finite; breakthrough occurs	Continuous; no exhaustion
Capital cost	Lower	High (generator, contactor, destructor)
Running cost	Carbon replacement and regeneration	Electricity — substantial
Side benefit	Removes odour as well	Raises biodegradability; disinfects
Main risk	Carbon disposal becomes a waste problem	Bromate formation; gas hazard

Table 2.7 Colour and COD removal: adsorption compared with oxidation.

Application to tannery wastewater

For a tannery, oxidation is used chiefly for COLOUR, not for disinfection — the effluent rarely requires the degree of pathogen control that would justify ozone on that ground alone. Ozone and the Fenton process are the realistic candidates, and the choice between them and activated carbon is an economic one made on the particular effluent. Where a biological stage follows, a partial oxidation step is often the most economical arrangement of all, because it converts refractory COD into food for the microbes.

2.7 Chemical Precipitation of Residual Metals

Purpose

To remove the small quantities of dissolved metals — principally chromium — that remain after the recovery unit described in Chapter 1.

Principle

Precipitation — the conversion of a dissolved substance into an insoluble solid by chemical reaction, so that it can afterwards be removed by settling or filtration.

Most metals are soluble as ions at low pH and insoluble as hydroxides at higher pH. Raising the pH therefore precipitates them:



Each metal has its own pH of minimum solubility, and this is the practical key to the process.

Metal	Optimum pH	Note
Chromium (III)	8.0 – 9.0	Redissolves above about 9.5 as chromite
Iron (III)	3.5 – 7.0	Very insoluble; precipitates readily
Zinc	9.0 – 10.0	Amphoteric; redissolves at high pH
Lead	9.0 – 9.5	Amphoteric
Copper	8.0 – 10.0	—

Table 2.8 Optimum pH for hydroxide precipitation of common metals.

Two consequences follow. First, since each metal has a different optimum, a mixed metal wastewater cannot be treated perfectly at any single pH; a compromise is chosen. Second, several metals are AMPHOTERIC — they redissolve if the pH is pushed too high — so overshooting the alkali dose actually worsens the result.

Process description

Alkali is dosed with rapid mixing to raise the pH into the optimum range; the mixture is then gently flocculated to grow the hydroxide particles; and finally the precipitate is separated by settling or filtration. A coagulant aid may be added, since metal hydroxide floc is light and gelatinous.

Sulphide precipitation is an alternative used where very low residual metal concentrations are required, because metal sulphides are far less soluble than the hydroxides. It is not used in tanneries for the obvious reason that adding sulphide to a tannery effluent creates the hazard that Chapter 1 was at pains to remove.

Important operating parameters

- pH: the single most important variable; must be held within the optimum band and monitored continuously.
- Mixing: rapid mixing for dosing, then gentle flocculation to grow the floc.
- Settling time: 1–2 hours; the hydroxide floc is light.
- Coagulant aid: 0.5–2 mg/L polyelectrolyte improves capture greatly.

Application to tannery wastewater

Where chrome recovery has been practised, only a small residual chromium load reaches this stage — typically well under 50 mg/L — and precipitation with lime or caustic within the coagulation step of Chapter 1 is usually sufficient to bring it below the 2.0 mg/L limit. Where segregation has NOT been practised, this stage becomes the only line of defence, the doses are much larger, and the resulting sludge is chrome-bearing hazardous waste. The contrast is a further illustration of the value of segregation at source.

2.8 Ion Exchange

Purpose

To remove specific dissolved ions selectively — metals, hardness, ammonium or particular anions — by exchanging them for harmless ions held on a solid resin.

Principle

Ion exchange — a reversible process in which ions held on a solid resin are exchanged for ions of the same charge present in the water.

The resin is a porous polymer bead carrying fixed charged groups, each balanced by a mobile counter-ion. When water passes through, ions in the water displace the counter-ions if the resin has a greater affinity for them. Two families exist:

Type	Fixed group	Exchanges
Cation exchange	Negatively charged (e.g. $-\text{SO}_3^-$)	Positive ions: Ca^{2+} , Mg^{2+} , Cr^{3+} , NH_4^+
Anion exchange	Positively charged (e.g. $-\text{N}^+(\text{CH}_3)_3$)	Negative ions: Cl^- , SO_4^{2-} , CrO_4^{2-} , NO_3^-

A typical exchange, softening water on a sodium-form cation resin, is written:



R denotes the resin matrix

Process description and regeneration

Water passes down through a packed bed of resin in a pressure vessel. As the resin loads, an exchange zone travels down the bed exactly as in a carbon column, until breakthrough occurs. The resin is then **REGENERATED**: a concentrated solution of the original counter-ion is passed through, reversing the exchange by mass action — brine (NaCl) for a softening resin, acid or alkali for demineralising resins.

The regeneration produces a small volume of highly concentrated waste containing all the removed ions. This stream is the principal disadvantage of the process: the pollutants have been concentrated rather than destroyed, and the concentrate must itself be treated.

Important operating parameters

Parameter	Typical value	Comment
Service flow rate	8 – 40 bed volumes per hour	Higher rates reduce capacity
Bed depth	0.8 – 2 m	—
Exchange capacity	1 – 2 eq/L of resin	Depends on resin type
Regenerant	NaCl, HCl, H_2SO_4 or NaOH	According to resin and duty
Feed suspended solids	< 5 mg/L	Solids foul the resin irreversibly
Feed organics	Low	Organic fouling is the commonest failure

Advantages and limitations

Advantages	Limitations
Highly selective — targets specific ions	Costly; resin is expensive and fouls
Achieves very low residual concentrations	Produces a concentrated regenerant waste
Resin is reusable for many cycles	Requires very clean feed water
Compact and reliable	Uneconomic for bulk desalination of a high-TDS water
Can recover a valuable ion for reuse	Organic fouling by tannins and dyes

Application to tannery wastewater

Ion exchange is NOT a practical answer to the tannery's bulk salinity problem. At 10,000–40,000 mg/L of TDS the resin would exhaust within minutes and the regenerant consumption would be absurd; reverse osmosis is the correct tool for that duty.

Its proper role is SELECTIVE polishing: recovering or removing residual chromium to reach a very low limit, removing ammonium, or softening water intended for reuse in a boiler. Applied to the right duty on a well-filtered water, it is precise and effective; applied to bulk desalination it is simply the wrong process.

2.9 Membrane Processes

Purpose

To separate dissolved and colloidal material from water by forcing the water through a selective barrier — and, uniquely among the processes in this material, to remove the dissolved salts.

2.9.1 The family of membranes

Membranes are classified by what they retain, and it is essential to know where each sits.

Process	Cut-off	Removes	Pressure
Microfiltration (MF)	0.1 – 10 μm	Suspended solids, bacteria	0.1 – 2 bar
Ultrafiltration (UF)	0.01 – 0.1 μm	Colloids, macromolecules, viruses	1 – 5 bar
Nanofiltration (NF)	$\approx 0.001 \mu\text{m}$	Divalent ions, colour, some salts	5 – 20 bar
Reverse osmosis (RO)	Non-porous	Essentially all dissolved salts	15 – 80 bar

Table 2.9 The membrane family, in order of increasing tightness.

Two points follow. First, MF and UF do NOT desalinate at all; their role in a reuse scheme is to protect the RO by removing everything that would foul it. Second, nanofiltration is a useful intermediate: it rejects divalent ions such as sulphate and the larger coloured molecules while passing much of the monovalent salt, at a fraction of the pressure RO requires.

2.9.2 Reverse osmosis: the principle

Place fresh water and salt water either side of a membrane that passes water but not salt. Water moves spontaneously into the salt solution, diluting it. This is OSMOSIS, and the pressure that would just stop the movement is the OSMOTIC PRESSURE of the salt solution.

Now apply to the salty side a pressure GREATER than the osmotic pressure. The flow reverses: water is squeezed out of the salt solution through the membrane, leaving the salt behind. This is REVERSE OSMOSIS.

It is important to understand that this is not filtration in the ordinary sense. The RO membrane has no pores that salt is too large to enter. Water dissolves into a dense polymer film and diffuses through it, while ions do not — a mechanism called SOLUTION–DIFFUSION. That is why the pressures required are so high and the energy cost so significant.

The osmotic pressure may be estimated for practical purposes by:

$$\pi \approx 0.077 \times \text{TDS} / 1000 \quad (\text{bar, with TDS in mg/L})$$

and the water flux through the membrane is driven by the excess of applied pressure over osmotic pressure:

$$J = A (\Delta P - \Delta \pi)$$

J = flux; A = membrane permeability; ΔP = applied pressure; $\Delta \pi$ = osmotic pressure difference

2.9.3 Recovery, rejection and the concentrate

Recovery — the fraction of the feed that emerges as clean permeate, typically 50–75% for a wastewater RO.

Rejection — the percentage of a dissolved species retained by the membrane; 95–99% for typical salts.

Concentrate (reject or brine) — the stream carrying the rejected salts, at correspondingly higher concentration.

$$\text{Recovery (\%)} = Q_p / Q_e \times 100$$

$$\text{Rejection (\%)} = (1 - C_p/C_e) \times 100$$

WORKED EXAMPLE 2.3 — Reverse osmosis mass balance

A tannery wishes to reuse its treated effluent. The feed to the RO is 400 m³/d at 12,000 mg/L TDS. The plant runs at 70% recovery with 97% salt rejection. Find the permeate and concentrate flows and qualities.

Step 1 Permeate flow $Q_p = 0.70 \times 400 = 280 \text{ m}^3/\text{d}$

Step 2 Concentrate flow $Q_c = 400 - 280 = 120 \text{ m}^3/\text{d}$

Step 3 Permeate TDS = $12,000 \times (1 - 0.97) = 360 \text{ mg/L}$ — well below the 2,100 limit; reusable

Step 4 Salt balance:

salt in feed = $400 \times 12,000 = 4,800,000 \text{ g/d}$

salt in permeate = $280 \times 360 = 100,800 \text{ g/d}$

salt in concentrate = $4,800,000 - 100,800 = 4,699,200 \text{ g/d}$

Step 5 Concentrate TDS = $4,699,200 / 120 = 39,160 \text{ mg/L}$

Step 6 Osmotic pressures: feed $\approx 0.077 \times 12.0 = 0.92 \text{ bar}$;

concentrate $\approx 0.077 \times 39.2 = 3.0 \text{ bar}$

Operating pressure must exceed π at the concentrate end: adopt 15–25 bar

SUMMARY: 280 m³/d of reusable water at 360 mg/L, and 120 m³/d of brine at 39,160 mg/L — nineteen times the discharge limit.

COMMENT: the plant has recovered 70% of its water, which was the object. But it has also produced a highly saline stream that cannot be discharged and cannot be treated biologically. That is the subject of Section 2.10.

2.9.4 Fouling: the operating problem

Every RO plant lives or dies by the control of fouling. Four kinds occur, and each has its own remedy.

Type	Cause	Control
Particulate fouling	Suspended matter depositing on the surface	MF/UF pre-treatment; filtration
Scaling	Sparingly soluble salts (CaSO ₄ , CaCO ₃) precipitating as they concentrate	Antiscalant dosing; limit recovery
Organic fouling	Organic matter adsorbing onto the membrane	Remove COD and colour beforehand (2.5, 2.6)
Biofouling	Bacterial growth forming a biofilm	Disinfect the feed — but note that free chlorine damages polyamide membranes and must be removed before the membrane

The universal lesson is that RO demands a thoroughly pre-treated feed. A typical reuse train therefore reads: treated effluent → filtration → carbon or ozone → ultrafiltration → antiscalant dosing → reverse osmosis.

Advantages and limitations

Advantages	Limitations
The only practical means of removing dissolved salt	High capital and energy cost (15–80 bar)
Produces water fit for reuse	Produces a concentrate that must be disposed of
Also removes colour, COD, metals and pathogens	Extensive pre-treatment essential
Modular — capacity easily extended	Membranes foul; chemical cleaning needed
Enables recovery of water and some chemicals	Membrane replacement every 3 – 7 years

KEY IDEA

Reverse osmosis does not destroy salt — it **CONCENTRATES** it. A plant recovering 70% of its water produces a brine roughly three times as saline as the feed, and that stream must then be dealt with. The concentrate is the central difficulty of every reuse scheme.

Application to tannery wastewater

Reverse osmosis is the only answer to the salinity problem identified in Section 2.1.1, and it is increasingly installed at large tanneries and common effluent treatment plants where water reuse or a TDS limit must be met. Its cost, both capital and running, is substantial, and it cannot be operated at all without the full pre-treatment train ahead of it. This is why reducing salt at source — low-salt or salt-free curing — is so often the better investment.

2.10 Evaporation, Crystallisation and Zero Liquid Discharge

Purpose

To deal with the concentrate produced by reverse osmosis, and — in the limiting case — to eliminate liquid discharge from the site altogether.

The problem

Worked Example 2.3 produced 120 m³/d of brine at nearly 40,000 mg/L. This stream cannot be discharged, cannot be treated biologically, and cannot be ignored. Four options exist.

Option	Assessment
Discharge to sea or deep well	Where geography and law permit — rarely available in Bangladesh
Solar evaporation ponds	Simple and cheap to run, but require large lined areas and are interrupted by rain
Mechanical evaporation and crystallisation	The zero-liquid-discharge route; complete but very costly
Reduce salt at source	The cheapest of all, and the only one that reduces the problem rather than relocating it

The ZLD train

Zero liquid discharge (ZLD) — a scheme in which no liquid effluent leaves the site: all the water is recovered for reuse and the dissolved salts end up as a solid.

1. Reverse osmosis recovers most of the water as clean permeate and concentrates the salts into a brine.
2. An EVAPORATOR — usually mechanical vapour recompression (MVR) or a multiple-effect evaporator — boils off further water, which is condensed and recovered. MVR reuses the latent heat of the vapour and so consumes 20–30 kWh per m³, against 600–700 kWh/m³ for simple thermal evaporation.
3. A CRYSTALLISER dries the remaining concentrate until the salts crystallise as a solid.
4. The recovered salt is reused where its purity permits, or landfilled as a controlled solid waste.

Advantages and limitations

Advantages	Limitations
No liquid discharge whatever	Very high capital cost
Near-total water recovery for reuse	Very high energy consumption
Addresses the salt problem completely	Complex operation requiring skilled staff
Satisfies the strictest regulation	Mixed salt recovered is often too impure to sell
Valuable where water is scarce or costly	Scaling and corrosion in the evaporator

KEY IDEA

Every kilogram of salt that enters the tannery must eventually leave it — in the effluent, in the RO concentrate, or as solid salt from a crystalliser. The cheapest point at which to deal with it is the curing yard, before it is ever dissolved.

Application to tannery wastewater

ZLD has been mandated for tannery clusters in parts of India and is under discussion elsewhere. It is technically complete and environmentally decisive, but it is the most expensive option in this material, and it is adopted where regulation compels it or where water scarcity makes recovery pay. For most Bangladeshi tanneries the realistic priorities remain, in order: reduce salt at source; recover chromium; treat properly; and consider membranes and ZLD only when the earlier measures have been exhausted.

2.11 Disinfection

Purpose

To destroy disease-causing micro-organisms before discharge or reuse.

Principle

Disinfection — the selective destruction of disease-causing organisms, sufficient to make the water safe. It is not sterilisation, which is the destruction of ALL organisms and is neither achievable nor necessary in wastewater treatment.

Disinfection is not instantaneous; organisms die progressively. The governing relation, due to Chick and Watson, is:

$$C^n \times t = \text{constant}$$

C = disinfectant concentration; t = contact time; n ≈ 1 for most disinfectants

With n taken as unity, the product $C \times t$ — the CT VALUE — becomes the single design quantity. A low dose held for a long time and a high dose held briefly achieve the same kill. This is why a disinfection unit is designed as a CONTACT TANK, and why short-circuiting in that tank is so damaging: water that crosses quickly receives only a fraction of the design CT.

Performance is expressed logarithmically: 1-log removal is 90%, 2-log is 99%, 3-log is 99.9%, 4-log is 99.99%. Moving from 99% to 99.99% sounds like a marginal gain but is in fact a hundredfold reduction in survivors.

The three methods

Feature	Chlorination	UV	Ozone
Nature	Chemical oxidant	Physical (radiation)	Chemical oxidant
Kills by	Enzyme inhibition	DNA damage	Cell-wall rupture
Residual	Yes — its unique merit	None	None
By-products	THMs, chloramines	None	Bromate if bromide present
Effect on colour	Slight	None	Strong
Cost	Lowest	Moderate	Highest
Needs clear water	Moderately	Critically	Moderately

Table 2.10 The three disinfection methods compared.

Chlorination

Chlorine hydrolyses in water to hypochlorous acid, which dissociates according to pH:



Both forms count as FREE AVAILABLE CHLORINE and appear as the same figure on a laboratory report, but they are not equivalent: hypochlorous acid is 40–80 times the more effective, because being uncharged it passes readily through the cell wall while the negatively charged hypochlorite ion is repelled by the negatively charged cell surface. Below pH 6 almost all the chlorine is HOCl; at pH 7.5 the two are equal; above pH 9 almost all is OCl[−]. Chlorination therefore works far better in slightly acidic water.

If ammonia is present — as it is in abundance in tannery effluent — chlorine reacts with it in preference to disinfecting, forming CHLORAMINES which are perhaps a hundred times weaker. Only when enough

chlorine has been added to destroy the chloramines does a free residual appear; that point is the BREAKPOINT, and it occurs at roughly 8–10 parts of chlorine to 1 part of ammonia nitrogen. This makes chlorination of tannery effluent expensive.

Ultraviolet treatment

UV light at 254 nm is absorbed by DNA, fusing adjacent thymine bases into dimers so that the strand can no longer be copied. The organism may remain metabolically alive but cannot reproduce, and an organism that cannot reproduce cannot infect. Nothing is added to the water and nothing is left behind.

The dose is the product of intensity and exposure time, in mJ/cm^2 , with 20–40 typical for wastewater disinfection. UV is particularly effective against protozoan cysts, which resist chlorine strongly.

Its limitation is decisive for tanneries: UV only works if the light reaches the organism. The measure is the UV TRANSMITTANCE, and a coloured tannery effluent may transmit only 20–30% against the 65% or more needed for economical operation. UV can therefore only be considered AFTER colour has been removed by carbon or oxidation — never before.

Ozone

Described in Section 2.6. As a disinfectant it is powerful and effective against all organism types, and it removes colour at the same time; it leaves no residual and no chlorinated by-products, but it is the most expensive of the three.

What interferes with disinfection

Factor	Effect
Suspended solids	Shield organisms and absorb UV. The most important interference of all
Organic matter	Consumes chlorine and ozone chemically, so less is available to act on organisms
Colour and turbidity	Absorb and scatter UV light
Ammonia	Converts chlorine to the far weaker chloramines
pH	Decides whether chlorine exists in the strong or the weak form
Temperature	Warmer water kills faster

Every one of these is present in tannery effluent, which is exactly why disinfection is placed LAST — on water that has already been filtered, decoloured and biologically treated.

Application to tannery wastewater

Disinfection is required where the receiving water is used for irrigation, bathing or fishing, or where the treated water is to be reused. For a tannery it is usually the least demanding of the requirements in this chapter, and chlorination is normally adequate and cheapest — though the ammonia content raises the dose considerably. Where the effluent is to be reused after RO, UV is commonly installed as a final barrier on the permeate, which is clear and therefore suits it well.

2.12 Distillation

Purpose and principle

Distillation separates a liquid mixture by exploiting the difference in volatility of its components. The liquid is boiled, the vapour is condensed, and the condensate — the distillate — is collected. Non-volatile substances such as salts remain behind.

Application in wastewater treatment

Distillation is rarely used for bulk wastewater treatment because boiling water is enormously energy-intensive. Its role in this field is limited to two situations:

- **Solvent recovery** — recovering organic solvents from a concentrated process stream, where the solvent's value justifies the energy. This has some application in finishing operations.
- **As the evaporation step in ZLD** — where the object is not purification but concentration of a small brine stream, and where mechanical vapour recompression makes the energy demand tolerable. This is Section 2.10.

For general tannery effluent, distillation is not an economic option and is mentioned here only for completeness.

2.13 Comparison of Secondary Technologies

Process	Removes	Mechanism	Cost	Residue
Filtration	Fine suspended solids	Straining, adhesion	Low	Backwash water
Activated carbon	Colour, refractory COD	Adsorption	Medium	Spent carbon
Ozone / AOP	Colour, refractory COD	Oxidation	High	None
Precipitation	Residual metals	Hydroxide formation	Low	Metal sludge
Ion exchange	Specific ions	Ion exchange	High	Regenerant brine
MF / UF	Colloids, bacteria	Size exclusion	Medium	Reject stream
NF / RO	Dissolved salts	Solution–diffusion	Very high	Concentrate
Evaporation / ZLD	Everything dissolved	Phase change	Extreme	Solid salt
Disinfection	Pathogens	Oxidation or DNA damage	Low–med	By-products (Cl ₂)

Table 2.11 Summary comparison of the technologies in this chapter.

Which process for which pollutant?

Residual pollutant	First choice	Alternative
Fine suspended solids	Granular filtration	Microfiltration
Colour	Activated carbon	Ozone or Fenton oxidation
Refractory COD	Activated carbon	Advanced oxidation
Residual chromium	Precipitation with the coagulation step	Ion exchange for a very low limit
Dissolved salts (TDS)	Reverse osmosis	Source reduction (preferable)
Pathogens	Chlorination	UV (needs clear water); ozone
RO concentrate	Evaporation and crystallisation	Solar ponds where land permits

2.14 Selecting a Treatment Train

The processes in this chapter are not applied as a fixed sequence. They are selected according to what the effluent must achieve, and four levels of requirement may be distinguished.

Requirement	What is needed	Typical train
Meet suspended-solids limit	Remove residual fine solids	Filtration only
Meet COD and colour limits	Remove refractory organics	Filtration → activated carbon (or oxidation)
Protect public health	Remove pathogens	Filtration → disinfection
Reuse the water	Remove dissolved salts	Filtration → carbon/ozone → UF → RO → UV
Zero liquid discharge	Eliminate liquid effluent	The above → evaporator → crystalliser

Table 2.12 Selection of an advanced treatment train according to requirement.

Two principles govern the choice, and both have appeared repeatedly in this chapter.

- **Do not install a unit that a cheaper one upstream would make unnecessary.** Removing colour biologically or by coagulation where possible reduces the carbon or ozone required; removing solids by filtration protects everything downstream.
- **Do not treat what need never have been created.** The salinity that makes RO and ZLD necessary comes from salt deliberately added in curing and pickling. Reducing it at source is almost always cheaper than removing it afterwards.

2.15 Chapter Summary

What this chapter covered

1. Primary treatment leaves behind refractory organics and colour, dissolved salts, residual fine solids and metals, and pathogens. This chapter dealt with all of these by non-biological means; the dissolved biodegradable fraction belongs to Chapter 3.
2. The units are placed in a fixed order — filtration, then adsorption or oxidation, then membranes, then disinfection — because each protects the next.
3. FILTRATION removes fine solids by straining, interception and adhesion. Dual media outlast single media because the coarse-to-fine gradation lets solids penetrate the whole bed.
4. ADSORPTION on activated carbon removes colour and refractory COD by holding molecules on an internal surface of 500–1,500 m²/g. Capacity is finite; breakthrough is inevitable; the pollution is transferred, not destroyed.
5. OXIDATION, by ozone or an advanced oxidation process, destroys the molecule instead. It costs more in energy but leaves no residue, and it raises the biodegradability of what remains — which can be exploited by placing it between two biological stages.
6. PRECIPITATION removes residual metals as hydroxides, each at its own optimum pH; several are amphoteric and redissolve if the pH is pushed too high.
7. ION EXCHANGE is precise and selective but is the wrong tool for bulk salinity; its place is polishing.
8. MEMBRANES are the only route to removing dissolved salts. Reverse osmosis does not destroy salt but concentrates it, and the concentrate is the central difficulty of every reuse scheme.
9. EVAPORATION and CRYSTALLISATION complete the picture as zero liquid discharge — technically complete, environmentally decisive and extremely costly.
10. DISINFECTION is governed by the CT value and is placed last, on clear water. Chlorination is cheapest and leaves a residual; UV adds nothing but demands clear water; ozone destroys colour as well.

The condition of the effluent after this chapter

Parameter	After primary treatment	After the processes in this chapter
Suspended solids	400 – 800 mg/L	< 10 – 30 mg/L
Colour	Strong	Within the 150 Hazen limit
Refractory COD	Present	Largely removed
Residual metals	Small soluble fraction	Below limits
Pathogens	Present	Removed
Dissolved solids	Unchanged from raw	Removed only if RO is installed
Dissolved biodegradable BOD	800 – 1,200 mg/L	Unchanged — see Chapter 3

Table 2.13 Effect of the processes in this chapter.

The last row is the point on which the whole material turns. None of the sophisticated processes in this chapter removes the dissolved biodegradable organic matter economically. Carbon would adsorb it, but would exhaust within hours; ozone would oxidise it, at a cost no plant could bear; membranes would reject

it, but foul immediately. That load is removed cheaply and reliably by one means only — living micro-organisms — and that is the subject of Chapter 3.

THE CENTRAL LESSON OF CHAPTER 2

These processes are precise, powerful and expensive. Each is the right answer to one particular residual pollutant, and the wrong answer to the others.

Use them selectively, place them in the correct order, and above all use them only on what genuinely requires them — after the cheap processes of Chapter 1 and the biological processes of Chapter 3 have removed everything they can.

Review questions

1. Name the pollutants that survive primary treatment, and state which are dealt with in this chapter and which in Chapter 3.
2. Why is filtration always placed first among the processes of this chapter? Give three reasons.
3. Explain, with reference to backwashing, why dual-media filters give longer runs than single-media sand filters.
4. Distinguish adsorption from absorption, and explain why activated carbon is so effective on dye molecules.
5. Write the Freundlich isotherm and explain the meaning of K_f and $1/n$.
6. A bench test removes COD from 350 to 120 mg/L using 0.8 g of carbon in 500 mL. Calculate the adsorption capacity q_e and the percentage removal.
7. What is breakthrough in a carbon column, and what must be done when it occurs?
8. Explain why ozone removes colour, in terms of what it does to a dye molecule.
9. Compare activated carbon with ozone under the headings mechanism, residue, capacity and cost.
10. Explain how a partial oxidation stage placed between two biological stages can treat what neither could handle alone.
11. Why does overshooting the alkali dose reduce the removal of an amphoteric metal?
12. Why is ion exchange unsuitable for removing the bulk salinity of tannery effluent?
13. Explain osmosis, reverse osmosis and osmotic pressure. Why is RO described as solution–diffusion rather than filtration?
14. An RO plant treats 300 m³/d at 10,000 mg/L TDS with 65% recovery and 96% rejection. Calculate the permeate and concentrate flows and concentrations by mass balance.
15. Name the four types of membrane fouling and the pre-treatment that controls each.
16. What is zero liquid discharge, and why is it so costly? What is the cheaper alternative?
17. State the Chick–Watson relation and explain the significance of the CT value.
18. Why does chlorination work better at pH 6 than at pH 9?
19. Why is UV disinfection difficult to apply to tannery effluent, and what must be done first?
20. A tannery must meet colour and COD limits but has no reuse requirement. Propose a treatment train and justify each unit.

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LOOKING AHEAD

Chapter 3 (Biological Treatment) takes up the one pollutant this chapter could not remove economically: the dissolved biodegradable organic matter that constitutes most of the remaining BOD.

It covers the aerobic suspended-growth processes — the activated sludge process and its modifications, together with the aeration systems that serve them; the attached-growth processes — trickling filters, rotating biological contactors and biofilters; the hybrid carrier systems; anaerobic treatment and biogas recovery; pond and lagoon systems; and the biological removal of nitrogen.